

W3(I) Connection mode.

If a 1D cell is vertically connected to the top of the 3D cell represented by this sdbvol, this number is 1.

If a 1D cell is vertically connected to the bottom of the 3D cell represented by this sdbvol, this number is -1.

If a 1D cell is horizontally connected to the 3D cell represented by this sdbvol, this number is 0.

Additional three words (W4, W5 and W6) should be entered in case of horizontal connection (i.e., W3 is 0).

W4(I) Gap number 1. This number is the gap number facing the 1D/3D interface surface represented by this sdbvol.

This number is zero if there is no gap facing the 1D/3D interface surface.

W5(I) Gap number 2. This number is the left gap number normal to the 1D/3D interface represented by this sdbvol.

This number is zero if there is no gap normal to the 1D/3D interface surface.

W6(I) Gap number 3. This number is the right gap number normal to the 1D/3D interface.

This number is zero if there is no gap normal to the 1D/3D interface surface. The sequence of W5 and W6 is meaningless.

8.21 Multi-Dimensional Input

A multi-dimensional component is indicated by MULTID on Card CCC0000. This component defines a one-, two-, or three-dimensional array of volumes and the internal junctions connecting the volumes. The multi-dimensional component is described as a three-dimensional component but can be reduced to two or one dimensions by defining only one interval in the appropriate coordinate directions. The geometry can be either Cartesian (x,y,z) or cylindrical (r,θ,z). In cylindrical geometry, the r-direction can start at zero or nonzero, and θ can cover 360 degrees (i.e., a full circle) or can cover less than 360 degrees (wedge shape, semicircle, etc.).

An orthogonal, three-dimensional grid is defined by mesh interval input data in each of the three coordinate directions. The edges of the hydrodynamic volumes are defined by the grid lines. Given n_x intervals in the x- or r-coordinate direction, n_y intervals in the y- or θ-coordinate direction, and n_z intervals in the z-coordinate direction, volumes are defined. The number of volumes in a threedimensional

component is limited to 9999 volumes. Volumes are numbered CCCXYYZZ0 where X, YY, and ZZ are the position numbers in the three coordinate directions. Position numbers in each coordinate

direction start with one at the origin and increase consecutively in the positive coordinate direction. X represents the position number for the first coordinate direction which is x in Cartesian geometry and r in cylindrical geometry. The use of one digit for the first coordinate limits the number of volumes in that coordinate to nine volumes. YY represents the position number of the second coordinate direction which is y in Cartesian geometry and θ in cylindrical geometry. ZZ represents the position number of the third coordinate which is z in both Cartesian and cylindrical geometries. The use of two digits for the second and third directions allows up to 99 volumes in those coordinate directions. The maximum values of X, YY, and ZZ are n_x , n_y , and n_z respectively. Positive θ direction is counterclockwise. The r- θ plane is in the same plane as the x-y plane.

The volume face number is given by CCCXYYZZF where the face number, F, is added to the volume number. The face numbers are 1 and 2 for the inlet and outlet faces respectively of the z-coordinate, 3 and 4 for the inlet and outlet faces for the x- or r- coordinate, and 5 and 6 for the inlet and outlet faces for the y- or θ - coordinate. The volume-face number is the volume connection code used in the from and to portion of junction input.

Junctions for this component are generated between all internal faces, that is all faces common to the volumes in the component. The number of junctions is $(n_x-1)*n_y*n_z + n_x*(n_y-1)*n_z + n_x * n_y*(n_z-1)$ for Cartesian geometry and for cylindrical geometry where the θ -coordinate does not cover a full circle. When the θ coordinate covers 360 degrees, the number of junctions increases by . The coordinate directions of the junctions are aligned in the positive directions of the coordinates. For numbering purposes, the junctions are associated with the from face of the two volumes being joined. Thus, the junctions are numbered CCCXYYZZF where F is limited to 2, 4, or 6.

External junctions may connect to any exterior faces of the volumes and also any interior faces.

External junctions connecting to internal faces imply branching or merging flow since internal junctions connect all internal faces. Some adjustments to the volumes and flow areas of the volumes and flow areas of the junctions should be made to account for the piping necessary to reach the internal face.

MULTID component can be connected to 1-D components externally via either a normal junction or a crossflow junction, depending on the actual flow paths. The 1-D to MULTID external junction connection to an external MULTID face should be restricted to 1 junction for each external MULTID face. The MULTID component can also be connected to MULTID components externally via either a normal junction or a crossflow junction. The MULTID to MULTID connection is restricted to the same direction (i.e., radial to radial, axial to axial, etc.).

8.21.1 Card CCC0001, Multi-Dimensional Information Card

This card is required.

W1(I)	Number of x or r coordinate intervals (nx). This word must be greater than zero and less than 10.
W2(I)	Number of y or θ coordinate intervals (ny). This word must be greater than zero and less than 100.
W3(I)	Number of z coordinate intervals (nz). This word must be greater than zero and less than 100.
W4(I)	Velocity/mass flow flag. This word is optional and if missing, is assumed to be zero. The initial junction conditions are velocities if zero is entered and are mass flows if one is entered.
W5(I)	Geometry and θ flag. This word is optional and if missing, is assumed to be zero. If this word is zero, Cartesian geometry is indicated; if the magnitude of this word is one, cylindrical geometry is indicated. If the sign of this word is positive, θ is assumed to extend to 360 degrees; if the sign of the word is negative, θ is assumed to extend to less than 360 degrees. Input checking uses $360 + 0.0005$ degrees for the region that represents 360 degrees. The extra plane of junctions perpendicular to the θ direction exists when this quantity is one and does not exist when this quantity is zero or minus one. The sign is appropriate only to cylindrical geometry.
W6(R)	Value of innermost radial coordinate (m, ft). This word is optional and if missing is assumed 0.0. This word must be zero in Cartesian geometry, and must be greater than or equal to zero in cylindrical or Cartesian geometry. A nonzero value allows the specification of a cylinder with a hollow center in cylindrical geometry. This word is not used in Cartesian geometry.
W7(I)	Three-dimensional flag. This word is optional and if missing is assumed to be zero. Not used yet, it has been reserved for future application.

8.21.2 Card CCC0002, Multi-Dimensional Rotation Angle Data Card

This card is optional and if missing, the angles for W1, W2, and W3 are assumed to be zero and W4 is dummy card.

W1(R)	First rotation angle (degrees).
W2(R)	Second rotation angle (degrees).
W3(R)	Third rotation angle (degrees).
W4(A)	dummy, not used.

The angles used in MULTID are Euler angles. The reference coordinates are the standard right hand x-, y-, z-coordinate system in Cartesian geometry. The z-axes of both Cartesian and cylindrical geometries coincide and are oriented in the vertical direction with the positive direction being upward. The x-, y-, r- and θ -coordinates are in the horizontal plane with the $\theta = 0$ line coinciding with the x-axis. The coordinate system of the multi-dimensional component is initially aligned with the reference coordinates. For the Euler angles, the first angle specifies the rotation of the component's coordinate system about its original z-axis, the second angle is the rotation of the component's coordinate system about its new x-axis, and the third angle is the rotation of the component's coordinate system about its new z-axis.

A positive angle specifies counterclockwise rotation as viewed by an observer on the positive part of the rotation axis looking towards the origin. In a 90 degree counterclockwise rotation about the z-axis, the position of the x axis would be moved to the previous position of the y-axis. Similarly, a 90 degree counterclockwise rotation about the x-axis would move the y-axis to the previous position of the z-axis. The input range of each Euler angle and pitch-yaw-roll angle is from 0.0 through plus or minus 360 degrees. The effects of the spherical first and second angles of the pipe and single volumes and the Euler and pitch-yaw-roll angles are similar, except that the spherical angle first angle is not used and the second angle is limited to 90 degrees. An example of usage of the Euler angle data is to move the axial coordinate in cylindrical geometry from the vertical direction to the horizontal plane.

8.21.3 Cards CCC0XNN, Multi-Dimensional Mesh Interval Cards

These required cards enter the mesh interval data for the three coordinate directions. The digit X is 1 for x- or r-coordinate, 2 for y or θ coordinate, and 3 for z-coordinate. The NN digits, which may range from 01 through 99, sequence the cards within a series and need not be consecutive. One or more sets of data in sequential expansion format may be entered on each card.

W1(R)	Mesh interval (m, ft, or degrees).
W2(I)	Volume coordinate number. The number of the first volume coordinate number next to the origin is 1 and the last volume number is nx for x-coordinate, ny for y-coordinate, or nz for z-coordinate. The last volume coordinate number must equal nx, ny, or nz depending on the value of X. In cylindrical geometry, the θ coordinates are entered in degrees. The sum of the intervals cannot exceed 360 degrees. The sum must equal 360 degrees if W5 on Card CCC0001 is 1 and must not equal 360 degrees if W5 is -1. Input checking uses 360 $\pm$ 0.0005 degrees for the region that represents 360 degrees. Presently a further

restriction exists on the q intervals when the innermost radius is zero. Then, the number of θ intervals must be even and the size of the interval must have 180 degree symmetry.

8.21.4 Cards CCC1NNN, Multi-Dimensional Volume Option Cards

These cards are required. The NNN digits range from 001 through 999 and need not be consecutive. These cards use an overlay format. The first six words define a rectangular solid in Cartesian geometry and an annular segment in cylindrical geometry. The solid consists of volumes where the x- or r-coordinate number ranges from x_1 through x_2 , the y or θ coordinate number ranges from y_1 through y_2 , and the z-coordinate number ranges from z_1 through z_2 . The quantities following the six overlay numbers apply to the volumes within the solid defined by the overlay. Overlays may reference any volume more than once; the data used in a volume is the information in the last overlay referencing that volume. Default data is stored in volumes not referenced by an overlay.

W1(I)	x_1
W2(I)	x_2
W3(I)	y_1
W4(I)	y_2
W5(I)	z_1
W6(I)	z_2
W7(R)	Volume porosity factor. Default is 1.0. This quantity must be greater than zero and less than or equal to one. The original volume of each volume is computed from the mesh interval data. The actual volumes's volume may be reduced by the factor in this word to account for solid material such as fuel pins within the volume. It could also account for piping which accesses interior volumes. The volume flow areas for the three coordinate directions are computed from the actual volume divided by the volume's length along the coordinate direction. The length is the appropriate mesh interval except for the θ direction where it is the mesh interval times the radius to the midpoint of the volume.
W8(I)	Scaler volume control flag. This word has the general packed format <u>tlpvbfe</u> but this word is limited to <u>tlpvb0e</u> . It is not necessary to input leading zeros. This word applies to the scaler oriented options for a volume.
W9(I)	X- or r-coordinate volume control flag. This word has the general packed format <u>tlpvbfe</u> but this word is limited to <u>00000f0</u> . It is not necessary to input leading zeros. This word applies to the coordinate options.

- W10(I) Y- or θ -coordinate volume control flag. This word has the general packed format tlpybfe but this word is limited to 00000f0. It is not necessary to input leading zeros. This word applies to the coordinate options.
- W11(I) Z-coordinate volume control flag. This word has the general packed format tlpybfe but this word is limited to 00000f0. It is not necessary to input leading zeros. This word applies to the coordinate options.

The digit t specifies whether the thermal front tracking model is to be used; t = 0 specifies that the front tracking model is not to be used for the volume, and t = 1 specifies that the front tracking model is to be used for the volume. The digit l specifies whether the mixture level tracking model is to be used; l = 0 specifies that the level model not be used for the volume, and l = 1 specifies that the level model be used for the volume. The digit p specifies whether the water packing scheme is to be used p = 0 specifies that the water packing scheme is to be used for that volume, and p = 1 specifies that the water packing scheme is not to be used. The digit v specifies whether the vertical stratification model is to be used. v = 0 specifies that the vertical stratification model is to be used, and v = 1 specifies that the vertical stratification model is not to be used. The digit b specifies the interphase friction to be used. b = 1 means that the Bestion/Analytis rod bundle interphase friction model is to be applied; b = 0 means that the normal interphase friction model is to be applied. The digit f specifies whether wall friction is to be computed. f = 0 specifies that wall friction effects are to be computed for the volume, and f = 1 specifies that wall friction effects are not to be computed. The digit e specifies if nonequilibrium or equilibrium is to be used. e = 0 specifies that a nonequilibrium (unequal temperature) calculation is to be used, and e = 1 specifies that an equilibrium (equal temperature) calculation is to be used. Equilibrium volumes should not be connected to nonequilibrium volumes. The equilibrium option is provided only for comparison to other codes. If this word is 2, all the water volume is in the film (i.e., no drops) when in the annular mist flow regime.

8.21.5 Cards CCC2NNN, Multi-Dimensional Volume Friction Data Cards

These cards are required. The NNN digits range from 001 through 999 and need not be consecutive. These cards use an overlay format similar to the CCC1NNN cards described above. Not all volumes need to be referenced by the overlay. At least 12 words must be entered on this card and 18 or 27 words may be entered. Words 13 through 18 enter additional wall friction data which specify laminar shape factors and viscosity ratio exponents. Words 19 through 27 specify data for an alternate turbulent friction factor model. The alternate turbulent data allow the specification of user defined friction factors for selected volumes and coordinate directions. The turbulent friction factor has the form $f = A + B(\text{Re})^{-C}$ where A, B, and C are entered for each coordinate. The standard turbulent model can be specified for a particular volume and coordinate by entering zeros for the three values. Continuation cards can be used to enter the large number of words for this card number.

W1(I)	x1
W2(I)	x2
W3(I)	y1
W4(I)	y2
W5(I)	z1
W6(I)	z2
W7(R)	Wall roughness for the x- or r-coordinate (m, ft). Default value is zero.
W8(R)	Hydraulic diameter for the x- or r-coordinate (m, ft). Default value is zero. If zero, the hydraulic diameter is computed from four times the total flow area in that direction divided by the associated wetted perimeter. The flow area and wetted perimeter are appropriate for the geometry and the coordinate direction. This represents the default for a tank.
W9(R)	Wall roughness for the y- or θ -coordinate (m, ft). Default value is zero.
W10(R)	Hydraulic diameter for the y- or theta-coordinate (m, ft). Default value is zero. If zero, the hydraulic diameter is computed from four times the total flow area in that direction divided by the associated wetted perimeter. The flow area and wetted perimeter are appropriate for the geometry and the coordinate direction. This represents the default for a tank.
W11(R)	Wall roughness for the z-coordinate (m, ft). Default value is zero.
W12(R)	Hydraulic diameter for the z-coordinate (m, ft). Default value is zero. If zero, the hydraulic diameter is computed from four times the total flow area in that direction divided by the associated wetted perimeter. The flow area and wetted perimeter are appropriate for the geometry and the coordinate direction. This represents the default for a tank.
W13(R)	Shape factor for x- or r-coordinate. Default value is 1.0.
W14(R)	Viscosity ratio exponent for x- or r-coordinate. Default value is 0.0.
W15(R)	Shape factor for y- or θ -coordinate. Default value is 1.0.
W16(R)	Viscosity ratio exponent for y- or θ -coordinate. Default value is 0.0.

W17(R)	Shape factor for z-coordinate. Default value is 1.0.
W18(R)	Viscosity ratio exponent for z-coordinate. Default value is 0.0.
W19(R)	A for x- or r-coordinate. Default value is zero.
W20(R)	B for x- or r-coordinate. Default value is zero.
W21(R)	C for x- or r-coordinate. Default value is zero.
W22(R)	A for y- or q-coordinate. Default value is zero.
W23(R)	B for y- or q-coordinate. Default value is zero.
W24(R)	C for y- or q-coordinate. Default value is zero.
W25(R)	A for z-coordinate. Default value is zero.
W26(R)	B for z-coordinate. Default value is zero.
W27(R)	C for z-coordinate. Default value is zero.

8.21.6 Cards CCC3001 Through CCC5999, Multi-Dimensional Junction Data and Initial Condition Cards

These cards are required. The range of card numbers need not be consecutive. These cards use the first six words as overlay information to specify a range of volumes and in addition use a seventh number to specify the volume face. The information following the face number is applied to the junction originating from the specified face of each volume included in the overlay. Not all junctions need to be referenced by these cards.

A totally blocked internal MULTID junction (i.e., the junction that is not defined in this card) is treated as a wall with no flow. If all the internal MULTID junctions associated with a MULTID volume are blocked (i.e., all the junctions are not defined), then that MULTID volume is treated as another system.

W1(I)	x1
W2(I)	x2
W3(I)	y1
W4(I)	y2
W5(I)	z1

W6(I)	z2
W7(A)	Face direction. The face direction is limited to x, y, or z.
W8(R)	Junction area factor. Default value is one. This quantity must be greater than or equal to zero and less than or equal to one. The original junction area is computed from the mesh interval data. The actual area is the original area times this factor. This quantity can be used to account for solid structures within the volume.
W9(R)	Reynolds number independent forward flow energy loss coefficient. Default value is zero.
W10(R)	Reynolds number independent reverse flow energy loss coefficient. Default value is zero.
W11(I)	Junction control flags. This word has the packed format <u>jefvcahs</u>. It is not necessary to input leading zeros. The digit j is not used and should be input as zero (j = 0). The jet junction model is not used. The digit e is not used and should be input as zero (e = 0). The digit f specifies CCFL options. f = 0 means that the CCFL model is not applied, and f = 1 means that it is applied. The digit v specifies stratification entrainment/pullthrough options. v = 0 means the model is not applied; v = 1 means an upward-oriented junction (offtake volume must be vertical); v = 2 means a downward-oriented junction (offtake volume must be vertical); and v = 3 means a centrally (side) located junctions. The digit c specifies choking options. c = 0 means that the choking model is applied, and c = 1 means that the choking model is not applied. The digit a specifies area change options. a = 0 means either a smooth area change or no area change, a = 1 means full abrupt area change model, (Kloss, area apportioning at branch, restricted junction area, and extra interphase drag), and a = 2 means a partial abrupt area change model (no Kloss, but includes area apportioning at branch, restricted junction area, and extra interphase drag). The digit h specifies nonhomogeneous or homogeneous. h = 0 specifies the nonhomogeneous (two-velocity momentum equations) option, and h = 1 or 2 specifies the homogeneous (single-velocity momentum equation) option. For the homogeneous option (h = 1 or 2), the major edit printout will show a 1.

The digit s specifies momentum flux option. $s = 0$ uses momentum flux in both the to and from volume. $s = 1$ uses momentum flux in the from volume, but not in the to volume. $s = 2$ uses momentum flux in the to volume but not in the from volume. $s = 3$ does not use momentum flux in either the to volume or the from.

- W12(R) Junction hydraulic diameter (m, ft). This word is optional. Default value is zero. If zero is entered, hydraulic diameter is set to four times the total junction area in that direction divided by the associated wetted perimeter. This represents the default for a tank.
- W13(R) Initial liquid velocity or mass flow (m/sec, ft/sec or kg/sec, lb/sec).
- W14(R) Initial vapor velocity or mass flow (m/sec, ft/sec or kg/sec, lb/sec).
- W15(R) This word is optional. If digit f of W11(I) is 1, this means CCFL data, β .
- W16(R) This word is optional. If digit f of W11(I) is 1, this means CCFL data, C_m .
- W17(R) This word is optional. If digit f of W11(I) is 1, this means CCFL data, C_0 .

8.21.7 Cards CCC6NNN, Multi-Dimensional Volume Initial Condition Cards

These cards are required. The NNN digits range from 001 through 999 and need not be consecutive. These cards use the first six words as overlay information to specify the range of volumes for which the following initial condition information applies. Each volume must be referenced at least once by these cards.

- W1(I) x1.
- W2(I) x2.
- W3(I) y1.
- W4(I) y2.
- W5(I) z1.
- W6(I) z2.
- W7(I) Control word. This word has the packed format ebt. It is not necessary to input leading zeros. The digit e specifies the fluid, where $e = 0$ is the default fluid. The value for $e > 0$ corresponds to the position number of the fluid type indicated on the 120 - 129 cards (i.e., $e = 1$ specifies H₂O, $e = 2$ specifies D₂O, etc.). The default fluid is that set for the hydrodynamic system by Cards 120 through 129 or this control word in another volume in

this hydrodynamic system. The fluid type set on Cards 120 through 129 or these control words must be consistent (i.e., not specify different fluids). If Cards 120 through 129 are not entered and all control words use the default $e = 0$, then H₂O is assumed as the fluid. The digit b specifies whether boron is present or not. The digit $b = 0$ specifies that the volume liquid does not contain boron; $b = 1$ specifies that a boron concentration in mass of boron per mass of liquid (which may be zero) is being entered after the other required thermodynamic information. The digit t specifies how the following words are to be used to determine the initial thermodynamic state. Entering $t = 0$ through 3 specifies only one component (steam/ water). Entering $t = 4$ through 6 allows the specification of two components (steam/water and noncondensable gas). With options t equal to 4 through 6, names of components of the noncondensable gas must be entered on Card 110, and mass fractions of the components are entered on Card 115.

If $t = 0$, the next four words are interpreted as pressure (Pa, lbf/in²), liquid specific internal energy (J/kg, Btu/lb), vapor specific internal energy (J/kg, Btu/lb), and vapor void fraction; these quantities will be interpreted as nonequilibrium or equilibrium conditions depending on the internal energies used to define the thermodynamic state. Enter only the minimum number of words required. If entered, boron concentration (mass of boron per mass of liquid) follows the last required word for thermodynamic conditions.

If $t = 1$, the next two words are interpreted as temperature (K, °F) and static quality in equilibrium condition. Enter only the minimum number of words required. If entered, boron concentration (mass of boron per mass of liquid) follows the last required word for thermodynamic conditions.

If $t = 2$, the next two words are interpreted as pressure (Pa, lbf/in²) and static quality in equilibrium condition. Enter only the minimum number of words required. If entered, boron concentration (mass of boron per mass of liquid) follows the last required word for thermodynamic conditions.

If $t = 3$, the next two words are interpreted as pressure (Pa, lbf/in²) and temperature (K, °F) in equilibrium condition. Enter only the minimum number of words required. If entered, boron concentration (mass of boron per mass of liquid) follows the last required word for thermodynamic conditions. The following options are used for input of noncondensable states only. In all cases, the criteria used for determining the range of values for static quality are: $1.0\text{E-}9 < \text{static quality} < 0.99999999$ implies two phase conditions; static quality $< 1.0\text{E}9$ or quality > 0.99999999 implies single phase. The static quality is given by $M_g / (M_g + M_f)$, where $M_g = M_s + M_n$. Noncondensable options are:

If $t = 4$, the next three words are interpreted as pressure (Pa, lbf/in²), temperature (K, °F), and static quality in equilibrium condition. Using this input option with static quality > 0.0 and < 1.0 , saturated noncondensables will result. Also, the temperature is restricted to be

less than the saturation temperature at the input pressure. Setting static quality to 0.0 is used as a flag that will initialize the volume to all noncondensable (dry noncondensable) with no temperature restrictions. Static quality is reset to 1.0 using this dry noncondensable option. Enter only the minimum number of words required. If entered, boron concentration (mass of boron per mass of liquid) follows the last required word for thermodynamic conditions.

If $t = 5$, the next three words are interpreted as temperature (K, °F), static quality, and noncondensable quality in equilibrium condition. Both the static and noncondensable qualities are restricted to be between $1.0\text{E-}9$ and 0.99999999 . Enter only the minimum number of words required. If entered, boron concentration (mass of boron per mass of liquid) follows the last required word for thermodynamic conditions. Little experience has been obtained using this option, and it has not been checked out.

If $t = 6$, the next five words are interpreted as pressure (Pa, lbf/in²), liquid specific internal energy (J/kg, Btu/lb), vapor specific internal energy (J/kg, Btu/lb), vapor void fraction, and noncondensable quality. These quantities will be interpreted as nonequilibrium or equilibrium conditions depending on the internal energies used to define the thermodynamic state. The combinations of vapor void fraction and noncondensable quality must be thermodynamically consistent. If noncondensable quality is set to 0.0, noncondensables are not present and the input processing branches to that type of processing ($t = 0$). If noncondensables are present (noncondensable quality greater than 0.0), then the vapor void fraction must also be greater than 0.0. If the noncondensable quality is set to 1.0 (pure noncondensable), then vapor void fraction must also be 1.0. When both the vapor void fraction and the noncondensable quality are set to 1.0, the volume temperature is calculated from the noncondensable energy equation using the input vapor specific internal energy. Enter only the minimum number of words required. If entered, boron concentration (mass of boron per mass of liquid) follows the last required word for thermodynamic conditions.

W8-W13(R) Quantities described under Word 7 (W7). Depending on the control word, two through five thermodynamic quantities may be required. Enter only the minimum required. If entered, boron concentration (mass of boron per mass of liquid) follows the last required word for thermodynamic conditions.

8.21.8 Cards CCC7001, Turbulent Mixing Length Cards

These cards are optional and if missing, the mixing length for W1 is set to be 0.1 m. The turbulent mixing length scale(m, ft) is used to calculate the turbulent viscosity for 1D module component CCC by Prandtl's mixing model.

W1(R) Enter the turbulent mixing length(m, ft). A real number should be required. if this card not exists, default 0.1(m, ft) is entered. **Table 8.21-1** suggests some useful guidelines to

calculate the quantity. The viscous fluid shear stresses are obtained by multiplying the velocity gradient and the sum of the dynamic viscosity and turbulent viscosity.

Table 8.21-1 Mixing length models for various flow conditions and geometries

Flow	turbulent length scale, l_m	hydraulic length scale, L
Mixing layer	$0.07L$	layer width
Jet	$0.09L$	jet half width
Wake	$0.16L$	wake half width
Axisymmetric jet	$0.075L$	jet half width
Boundary layer	$0.09L$	layer thickness
Channel	$L[0.14 - 0.08(1 - y/L)^2 - 0.06(1 - y/L)^4]$	channel half width

8.21.9 Cards CCC8NNN, Output Control Cards

These cards are optional. The NNN digits range from 001 through 999 and need not be consecutive. These cards use the first six words as overlay information to specify the range of volumes for major output activation. MULTID junctions between activated volumes also appears in the major output.

W1(I) x1.

W2(I) x2.

W3(I) y1.

W4(I) y2.

W5(I) z1.

W6(I) z2.

W7(I) control word. This word is dummy value and reserved for future use. enter 0.